

What we're doing

Can we curate at least **3,000,000** usable predicted protein-complex contacts-v1 training documents from the NVIDIA/AlphaFold Database complex release, with relaxed but calibrated confidence tiers, without allowing a handful of repeated interfaces to dominate the training mixture?

Why

The NVIDIA/AFDB complex release already contains enough usable predictions to reach three million training documents without running AlphaFold-Multimer ourselves. The earlier [#145](<https://github.com/Open-Athena/MarinFold/issues/145>) attempt became small because it was designed as a validation set: it restricted candidates to homologues of validation monomers and then selected one representative per coarse pair of independently clustered chains. For training, retaining multiple documents per **interface-aware** cluster while controlling their sampling weight should preserve the desired scale and prevent extreme redundancy.

We expect all usable high-confidence predictions to provide the core corpus. A calibrated medium-confidence tier should fill the remainder to three million. Homodimers and heterodimers, confidence tiers, and source datasets must remain separately identifiable so later training experiments can tune or ablate the mixture.

Results so far — the live census landed

The full 8.45 GB metadata mirror is normalized and censused: ****29,025,020 models**** (21,430,663 homodimers, 7,594,357 heterodimers), reproducing #145's dimer count exactly. ****1,923,625**** clear both the hard validity filter and the Tier-A quality gate.

Stage A itself is implemented and tested: lossless normalization, the yield census, exact sequence annotations with eval decontamination, deterministic Tier A/B quota selection, and a per-model terminal-status ledger.

Two findings that change the plan

****The `pilot-v0` floor of 0.5 cannot reach 3M.**** It yields 2,946,613 candidates **before** sequence-pair dedup and decontamination. The target needs a floor at or below ~ 0.4 , and realistically ~ 0.3 to leave decontamination margin. Whether that is defensible is the Stage-B pilot's job to decide.

****The 500k heterodimer floor is unreachable from AFCDB.**** Only 922,359 hard-eligible heterodimers have a nonzero ipSAE at all, and 353,774 reach $\text{ipSAE} \geq 0.1$. Clearing 500k would mean accepting models with essentially no predicted interface. The realistic ceiling is $\sim 210\text{k}$ at floor 0.30. The issue already conditioned this floor on the calibrated yield supporting it; it does not.

Archive reconnaissance: Stage D must move to the cloud

The coordinate archive is **48.8 TB** (35.20 TB homodimers, 13.58 TB heterodimers) across 17,820 referenced tars. Measured single-stream EBI throughput from the workstation is ~113 KB/s, so no variant of this runs locally.

Two properties make a selective extraction viable instead of a bulk mirror: each model is one member named `AF-<modelEntityId>-model\_v1.cif.zst`, so member names are a pure function of the model id; and EBI serves HTTP range requests, so tar headers can be walked with 512-byte reads and only selected members fetched. That moves roughly the selected payload (~1.5-2.5 TB) rather than 48.8 TB. Homodimer `chunk\_\*` tars also carry a redundant `.pdb.zst` per model that selective extraction skips entirely.

The sequence dependency, resolved

AFCDB ships no FASTA and ``modelEntityId`` is opaque, so subunit sequences come from AlphaFold DB's bulk ``sequences.fasta`` — measured at ****118.0 GB****, not the 110 GB the listing rounds to. A smoke run in ``europe-west4-a`` measured ****14.5 MB/s**** on one 2-vCPU pod, and the same rate with 4 ranges as with 16, so the scan is Python-bound rather than network-bound. It is therefore sharded across 16 pods, each owning a byte window and subdividing it into ranges.

A sharded run cannot judge its own completeness, so ``--verify`` checks the union: that the windows tile the file contiguously and that every requested accession was matched exactly once.

Three bugs the scale found

****The header parser.**** AFDB writes `>AFDB:AF-<acc>-F1 ... UA=<acc>`, and the leading `AFDB:` defeated the `AF-...-F<n>` pattern, so every header would have produced the whole token as the "accession" and failed the missing-accession check for every row.

****EBI refuses concurrent connections.**** Seven of the first sixteen pods died on `Connection refused` from the opening HEAD. The reader now retries with backoff and jitter and resumes a dropped stream from the first unreturned byte.

****A truncated transfer reads as a clean EOF.**** Writing the reconnect test found something worse than the bug it was written for: when a server closes early, `read()` returns `b""`, so an interrupted transfer was indistinguishable from the end of the file. The range would stop short, report success, and ship a short sequence set — under-decontaminating the corpus, silently, weeks downstream. Ranges now carry the source length and treat running dry before the range end as fatal.

Guarding the join

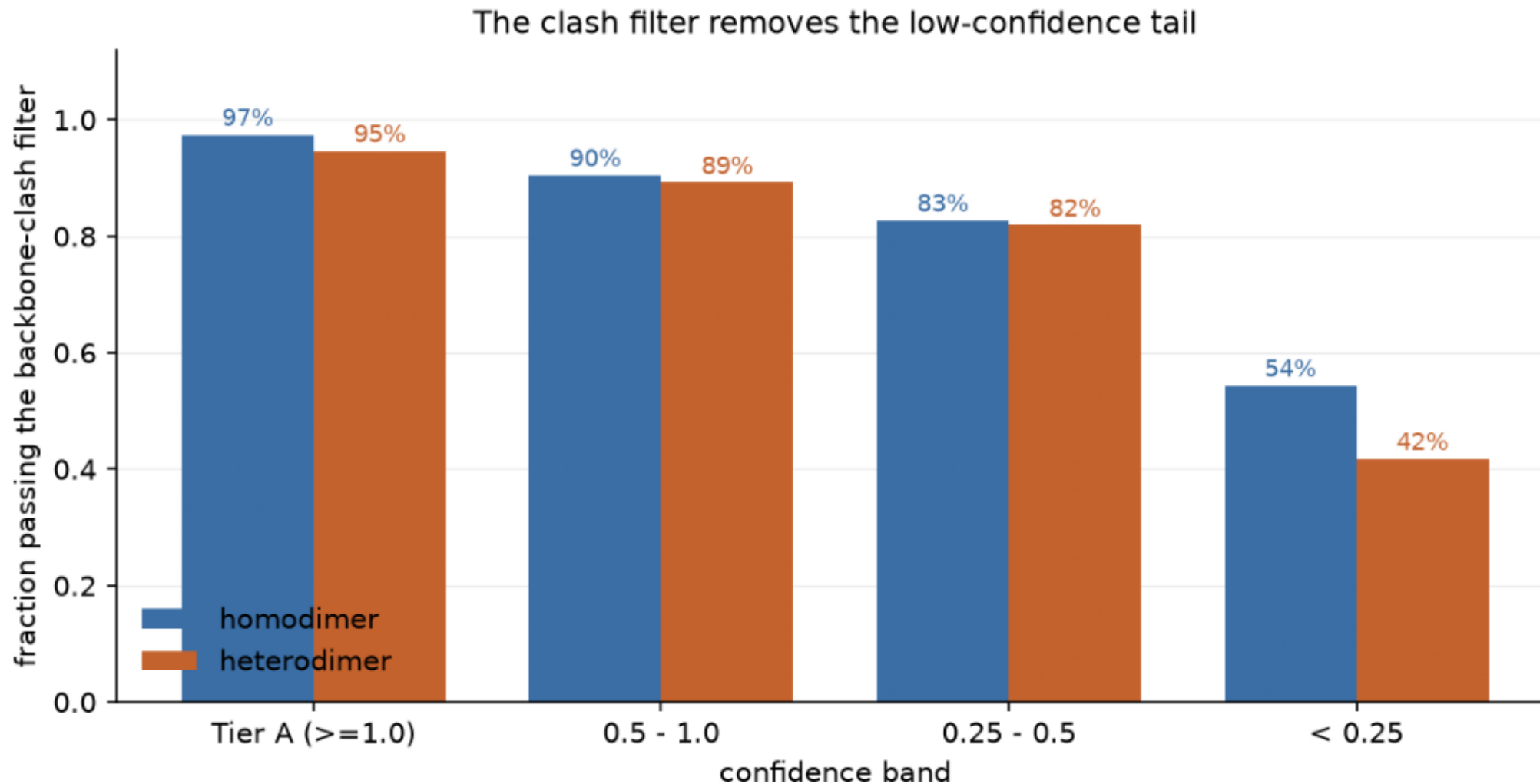
The selector now rejects any model whose joined UniProt subunit lengths disagree with the modelled residue count `n0chn`. Every AFCDB chain is at most 1,500 residues, well under AFDB's 2,700-residue fragmenting threshold, so an `-F1` model covers residues 1..L and the two should agree exactly. When they do not, the sequence we hashed, deduplicated and decontaminated on is not the sequence the document will be generated from, and the eval-leakage decision for that row is unsound.

Still to come

The eval2-v1 reference (577 = #225's 554 + #226's 23) is assembled and the drop-list builder is tested end to end against mmseqs. Remaining: run the decontamination search over the fetched subunits, run the selector, draw the 50k pilot, and only then design Stage D — where the reconnaissance already says range-addressed extraction moves ~1.5–2.5 TB instead of the full 48.8 TB.

clash_filter_by_band.png

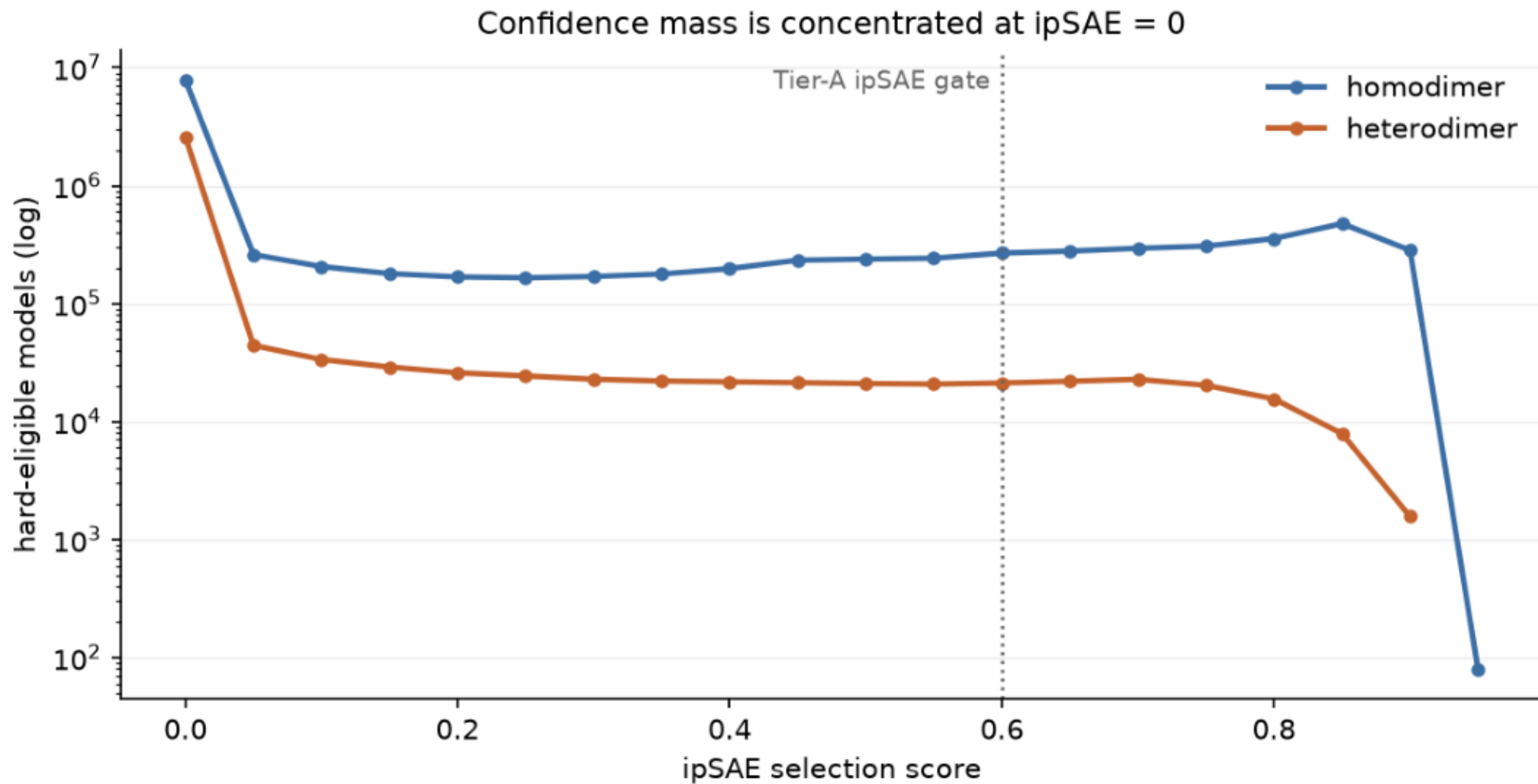
Backbone-clash pass rate by confidence band. The filter is quality-correlated rather than mis-calibrated: it keeps ~95-97% of Tier A and rejects most of the weakest band.



```
$ plots.py --input /data/exp294_predicted_complexes/metadata/normalized_*.parquet
```

ipsae_distribution.png

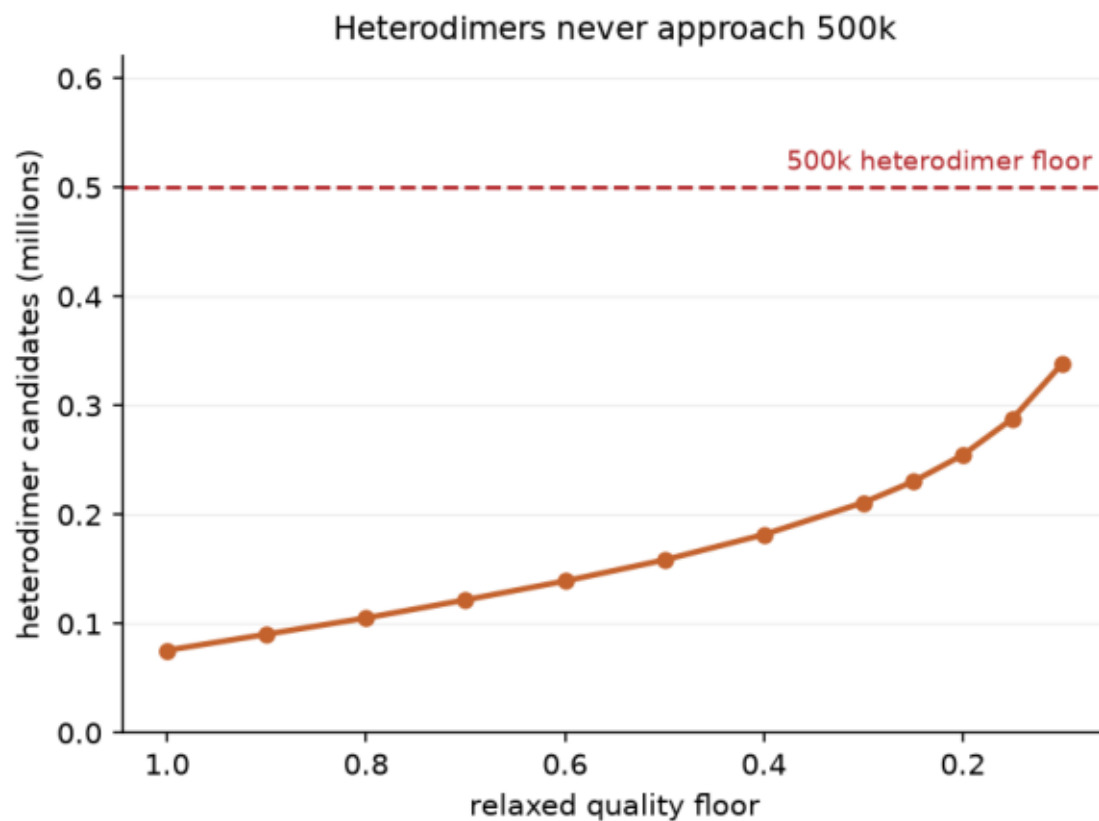
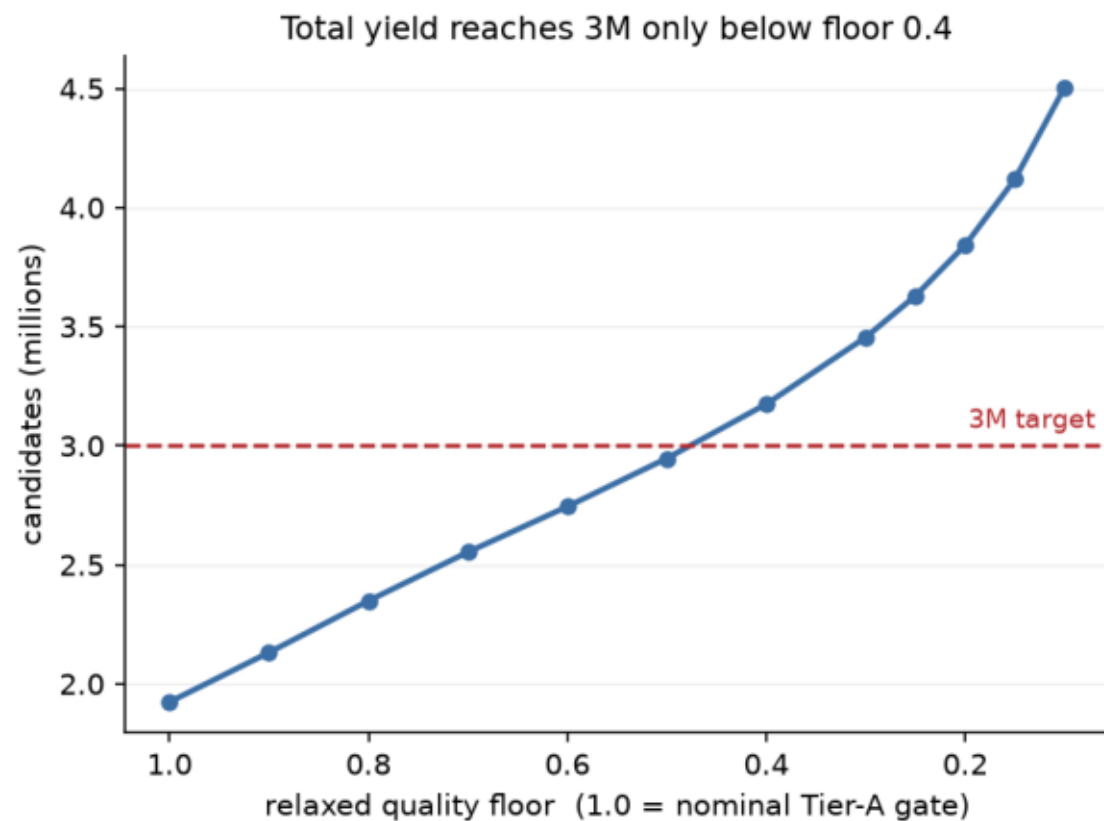
ipSAE over hard-eligible models (min over directions for homodimers, max for heterodimers; the two differ by 0.0014 on average).
76% of heterodimers score exactly 0, which is why relaxing the floor buys so little.



```
$ plots.py --input /data/exp294_predicted_complexes/metadata/normalized_*.parquet
```

yield_frontier.png

Hard-eligible AFCDDB candidates vs the relaxed quality floor. The 3M target needs a floor at or below ~0.4; 500k heterodimers is unreachable at any floor.



```
$ plots.py --input /data/exp294_predicted_complexes/metadata/normalized_*.parquet
```